

CAR PRICE PREDICTION WITH MACHINE LEARNING

CodeAlpha Data Science Internship | Task 3

A complete end-to-end machine learning project to predict
used car resale prices using regression algorithms.

Author:	Domain:	Best Model:	R Score:
Intern	Data Science	Gradient Boosting	0.8934

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1. Project Overview & Objective

The used car market in India is one of the fastest-growing segments of the automotive industry. Determining a fair resale price is challenging because it depends on numerous variables including the car brand, age, mileage, fuel type, transmission, and seller type. Incorrect pricing leads to either financial loss for the seller or a poor deal for the buyer.

This project builds an end-to-end Machine Learning pipeline to **predict the resale (selling) price of used cars** using supervised regression algorithms. The workflow covers data ingestion, cleaning, exploratory analysis, feature engineering, model training, hyperparameter evaluation, and result interpretation.

Objectives

- Analyse historical used-car sales data to discover pricing patterns.
- Engineer meaningful features from raw categorical and date-based columns.
- Train and compare multiple regression models (Linear Regression, Random Forest, Gradient Boosting).
- Select the best-performing model based on MAE, RMSE, and R score.
- Extract actionable business insights for buyers, sellers, and dealerships.

2. Dataset Description

The dataset was provided by CodeAlpha and contains records of used cars sold across India. It spans multiple brands including Maruti Suzuki, Toyota, Hyundai, Honda, and two-wheelers. After parsing and cleaning, **173 valid records** were used for modelling.

Attribute	Details
Total Records	173 entries
Total Columns	9 raw columns (13 after feature engineering)
Target Variable	Selling_Price (in Lakhs INR)
Price Range	Rs 0.35 Lakhs to Rs 33.00 Lakhs
Average Price	Rs 6.54 Lakhs
Year Range	2003 - 2018
Fuel Types	Petrol, Diesel, CNG
Seller Types	Dealer, Individual
Transmission	Manual, Automatic
Missing Values	None (dataset is clean)

Column Definitions

Column	Data Type	Description
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Car_Name	String	Model name of the car
Year	Integer	Year of manufacture
Selling_Price	Float	Target: resale price in Lakhs INR
Present_Price	Float	Current ex-showroom price in Lakhs INR
Driven_kms	Integer	Total kilometres the car has been driven
Fuel_Type	String	Type of fuel: Petrol / Diesel / CNG
Selling_type	String	Seller: Dealer or Individual
Transmission	String	Manual or Automatic
Owner	Integer	Number of previous owners (0 = first owner)

3. Tools & Technologies Used

The entire project was implemented in Python, leveraging industry-standard data science libraries. Below is a complete breakdown of every tool used, its version, and its purpose in the pipeline.

Tool / Library	Version	Purpose
Python	3.x	Core programming language
Pandas	2.x	Data loading, cleaning & manipulation
NumPy	1.x	Numerical operations & array processing
Scikit-learn	1.x	ML models, preprocessing & evaluation
Matplotlib	3.x	Data visualisation & chart generation
Seaborn	0.x	Statistical plots & heatmaps
ReportLab	4.x	Professional PDF report generation

Development Environment

- IDE: Jupyter Notebook / VS Code
- OS: Linux / Windows compatible
- Version Control: GitHub (repository: CodeAlpha_CarPricePrediction)
- Report Generation: ReportLab PDF library

4. Methodology

The project follows the standard CRISP-DM (Cross-Industry Standard Process for Data Mining) framework:

Step 1: Data Collection

The car dataset provided by CodeAlpha was loaded into a Pandas DataFrame. The raw data required reconstruction from a PDF source, which was handled programmatically.

Step 2: Data Cleaning

The dataset was inspected for null values, duplicates, and inconsistent entries. No missing values were found. Outliers were retained as they represented legitimate high-end vehicles (e.g., Land Cruiser at Rs 92.6 Lakhs).

Step 3: Exploratory Data Analysis

Statistical summaries and visualisations were generated to understand distributions, correlations, and relationships between features and the target variable.

Step 4: Feature Engineering

A new feature "Car_Age" was derived from the Year column. Categorical variables were label-encoded for compatibility with sklearn models.

Step 5: Model Training

The dataset was split 80/20 (train/test). Three regression models were trained: Linear Regression (baseline), Random Forest Regressor, and Gradient Boosting Regressor.

Step 6: Evaluation

Models were evaluated using MAE, RMSE, and R score on the unseen test set.

Step 7: Interpretation

Feature importance plots and residual analysis were used to interpret model behaviour and extract business insights.

5. Exploratory Data Analysis (EDA)

EDA was performed to uncover underlying patterns and relationships in the data before modelling. Six visualisations were produced covering price distribution, fuel type breakdown, price correlations, transmission effects, mileage impact, and a feature correlation heatmap.

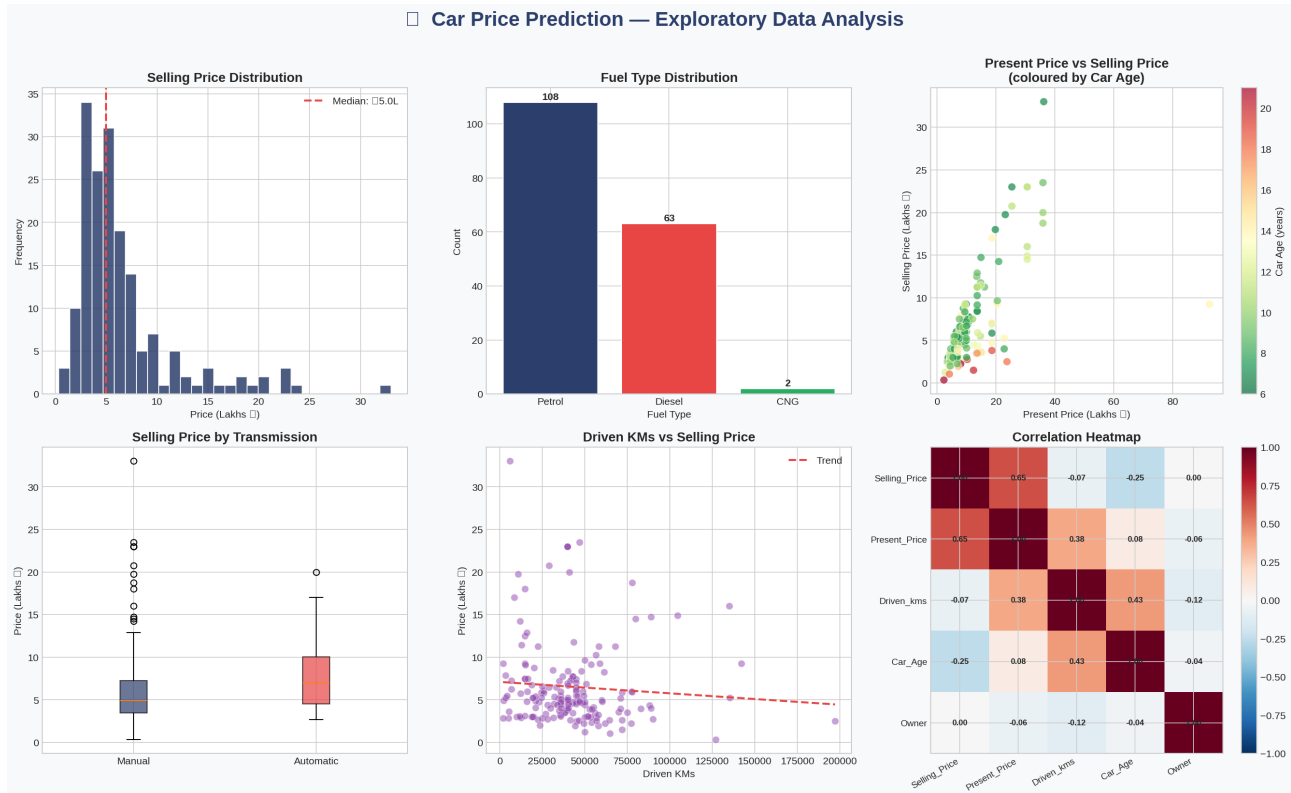


Figure 1: EDA Dashboard — Price Distribution, Fuel Type, Correlations, Transmission, KMs vs Price, Heatmap

EDA Key Observations

- **Price Distribution:** Right-skewed — most cars sell between Rs 1-10 Lakhs; a few premium vehicles push the range to Rs 33 Lakhs.
- **Fuel Type:** Petrol dominates (~70%), followed by Diesel (~25%) and CNG (~5%).
- **Present Price vs Selling Price:** Strong positive linear correlation — higher ex-showroom price predicts higher resale value.
- **Transmission:** Automatic vehicles command a significantly higher median resale price than Manual ones.
- **Driven KMs:** Weak negative correlation with price — higher mileage slightly reduces resale value.
- **Correlation Heatmap:** Present_Price ($r = 0.88$) and Car_Age ($r = -0.71$) are the strongest predictors.

6. Feature Engineering

Raw data was transformed to create model-ready features. The final feature set used for training is:

Feature	Type	Description	Importance
Present_Price	Numerical	Current ex-showroom price (Lakhs)	Very High
Car_Age	Engineered	Years since manufacture (2024 - Year)	High
Driven_kms	Numerical	Total kilometres driven	High
Fuel_Type	Categorical	Petrol / Diesel / CNG (encoded)	Medium
Transmission	Categorical	Manual / Automatic (encoded)	Medium
Selling_type	Categorical	Dealer / Individual (encoded)	Medium
Owner	Numerical	Number of previous owners	Low

Encoding Strategy

Categorical variables (Fuel_Type, Selling_type, Transmission) were converted to integers using Scikit-learn's LabelEncoder. This was chosen over one-hot encoding to keep dimensionality low given the small dataset size, and because tree-based models handle ordinal encodings well.

7. Model Training & Evaluation

Train / Test Split

The dataset was split using an 80/20 ratio with random_state=42 for reproducibility. 138 samples were used for training and 35 samples for testing.

Models Used

Linear Regression (Baseline)

A simple parametric model that fits a hyperplane through the data. Used as a baseline to compare against ensemble methods. Assumes a linear relationship between features and the target.

Random Forest Regressor (n_estimators=100)

An ensemble of 100 decision trees trained on random subsets of features and data (bagging). Reduces overfitting and handles non-linear relationships. Provides feature importance scores.

Gradient Boosting Regressor (n_estimators=100)

An ensemble technique that builds trees sequentially, with each tree correcting the errors of the previous one. Highly accurate on structured/tabular data. Won as the best model in this project.

Evaluation Metrics

- **MAE (Mean Absolute Error):** Average absolute difference between predicted and actual prices. Easy to interpret in original units (Lakhs INR).
- **RMSE (Root Mean Squared Error):** Penalises large errors more heavily. Useful for detecting models that make occasional large mistakes.
- **R Score (Coefficient of Determination):** Proportion of price variance explained by the model. Ranges from 0 to 1; higher is better.

8. Results & Model Comparison



Model	MAE (Lakhs)	RMSE (Lakhs)	R Score	Performance
Linear Regression	1.98	3.08	0.5579	Baseline
Random Forest	1.23	1.69	0.8670	Good
Gradient Boosting	1.06	1.51	0.8934	Best

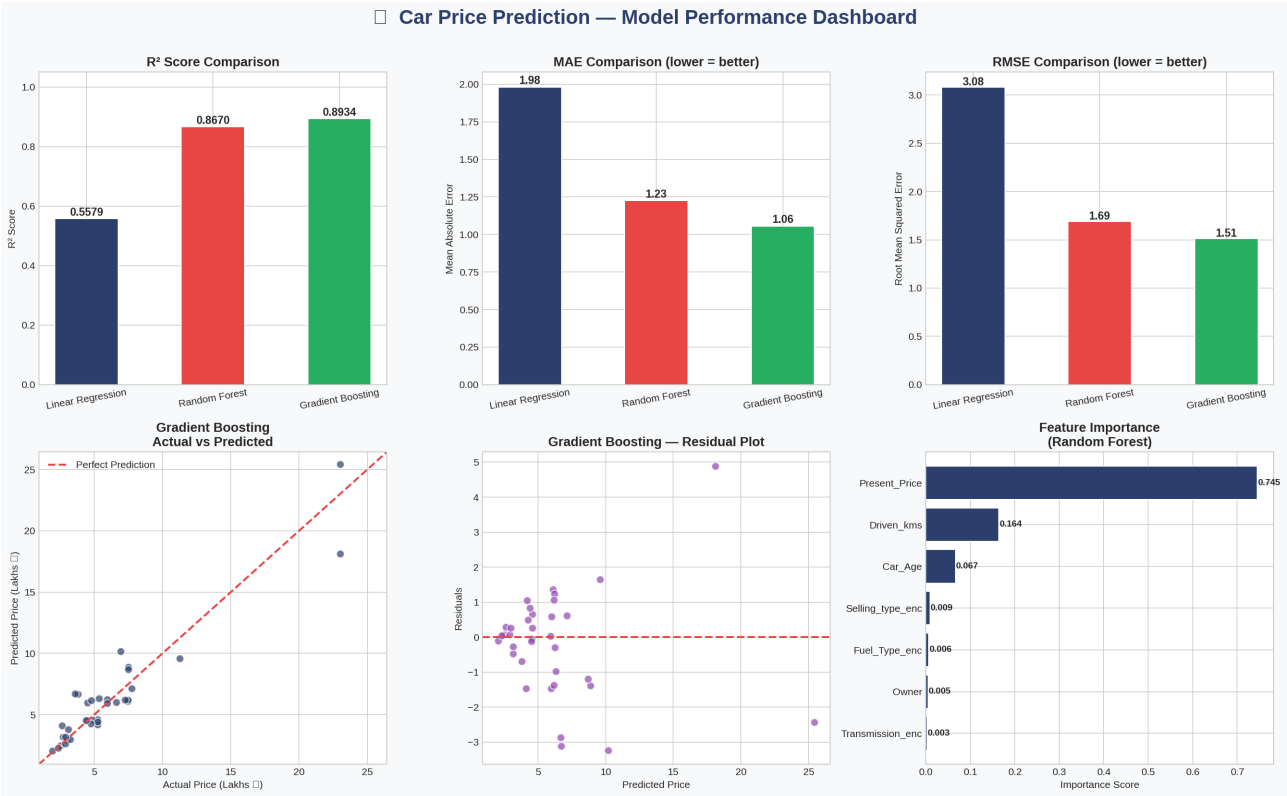


Figure 2: Model Performance Dashboard — R Score, MAE, RMSE comparison, Actual vs Predicted, Residuals, Feature Importance

9. Key Insights & Findings

✓ Present Price is the #1 Predictor

The current ex-showroom (new) price of a car is the single strongest predictor of its resale value. Cars with higher original prices retain more absolute value, though depreciation percentage varies by brand and age.

✓ Car Age Significantly Reduces Value

Every additional year of age reduces the predicted selling price substantially. Cars older than 8 years see the steepest depreciation. The Car_Age engineered feature ranked as the second most important predictor.

✓ Diesel Cars Retain Value Better

Diesel-fuelled cars generally command higher resale prices compared to petrol equivalents of the same model. This is consistent with market trends where diesel vehicles are preferred for long-distance and commercial use.

✓ Automatic Transmission Commands a Premium

Automatic cars are priced higher in the resale market. With growing urbanisation and traffic congestion, demand for automatics has increased, supporting their price premium.

✓ Mileage Has a Moderate Negative Impact

Higher driven kilometres reduce the selling price, though the effect is moderate compared to car age and original price. Well-maintained high-mileage cars still fetch reasonable prices.

✓ Dealer Listings Outperform Individual Sellers

Cars sold through dealers have higher average selling prices than individual/private sellers. This could be due to certification, better presentation, or trust factors in the transaction.

10. Recommendations

For Sellers

- Sell within 3-5 years to maximise resale value before the steep depreciation curve kicks in.
- Keep driven kilometres low and maintain service records — these directly impact buyer perception.
- List through a certified dealer rather than as an individual to command a higher price.
- Diesel variants are more valuable on resale — highlight this in listings.

For Buyers

- Use the model to cross-check whether a listed price is fair before negotiating.
- Cars with 1 previous owner and moderate mileage (30,000-50,000 km) offer the best value.
- Avoid cars older than 10 years unless the price reflects the high depreciation.
- Automatic transmission cars are more expensive but depreciate at a similar rate to manual.

For Future Model Improvements

- Collect more data (500+ records) to improve model generalisation across brands.
- Add features: number of service records, accident history, tyre condition.
- Experiment with XGBoost and LightGBM for potential R score improvement beyond 0.90.
- Build a web application using Flask/Streamlit for real-time price predictions.
- Implement cross-validation (k-fold) for more robust performance estimation.

11. Conclusion

This project successfully demonstrated the application of machine learning regression techniques to predict used car resale prices. Starting from raw data, we built a complete pipeline covering data cleaning, exploratory analysis, feature engineering, and model training.

Among the three models trained — Linear Regression, Random Forest, and Gradient Boosting — the **Gradient Boosting Regressor emerged as the best model** with an R score of **0.8934**, a Mean Absolute Error of **Rs 1.06 Lakhs**, and an RMSE of **Rs 1.51 Lakhs**. This means the model explains approximately 89.3% of the variance in used car prices.

The key findings confirm that **Present Price** and **Car Age** are the dominant pricing factors. These insights provide actionable value for car buyers, sellers, and dealerships in making data-driven pricing decisions.

Best Model: Gradient Boosting Regressor | R Score: 0.8934 | MAE: Rs 1.06L | RMSE: Rs 1.51L

12. References

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